

Determination of Mean Excitation Energies of Gaseous Helium, Nitrogen, and Argon from Alpha-Particle Energy Loss

Amir Assadi, Toby Matthews

*Department of Physics and Astronomy, University of Manchester
Oxford Road, Manchester M13 9PL, United Kingdom*

(Dated: April 26, 2026)

Mean excitation energies (I -values) are a fundamental input to stopping power calculations, entering directly into theoretical models such as the Bethe formula and playing a central role in particle transport simulations, detector calibration, and energy loss modelling. Accurate values are particularly important for gases such as helium, nitrogen, and argon, which are widely used in radiation detection, gas-based detectors, and plasma-related applications. In this work, the energy loss of alpha particles was measured as they traversed these gases at varying pressures, with energy spectra recorded using a silicon detector. The spectra were analysed to extract peak positions corresponding to different gas thicknesses, from which discrete stopping power values were determined. These were then used to compute corresponding I -values, yielding 41.1 ± 11.4 eV for helium, 79.8 ± 11.4 eV for nitrogen, and 178.4 ± 23.4 eV for argon.

1. INTRODUCTION

Mean excitation energies describe the average energy required to excite electrons in a material during charged particle interactions. They enter directly into the Bethe stopping power formalism, which governs the energy loss of fast charged particles in matter [1]. Because stopping power determines how particles deposit energy along their path, accurate I -values are essential for modelling radiation transport and detector response.

These quantities are widely used in particle and medical physics, particularly in detector calibration and radiation transport simulations. Any uncertainty in the mean excitation energy propagates into predictions of energy loss, making it a key parameter in both experimental and computational studies.

Despite their importance, I -values are not directly measurable and must be inferred from experimental energy loss data or theoretical calculations [2]. Gaseous systems such as helium, nitrogen, and argon are commonly used in detector environments and calibration setups. Alpha particles provide a well-defined probe due to their discrete energies and predictable interaction behaviour in matter.

This experiment determines the mean excitation energies of helium, nitrogen, and argon by measuring the energy loss of alpha particles at varying gas pressures. Stopping power is extracted from recorded energy spectra and used to infer I -values across different gas thicknesses.

2. THEORETICAL FRAMEWORK

The energy loss of charged particles in matter is described by the Bethe-Bloch equation, which provides a theoretical expression for the stopping power [1, 2]. In the non-relativistic regime, appropriate for the energies considered in this work, it is given by

$$-\frac{dE}{dx} = \frac{4\pi z^2 e^4}{m_e v^2} \frac{Z\rho}{A} \ln \left(\frac{2m_e v^2}{I} \right), \quad (1)$$

where z and v are the charge and velocity of the incident particle, ρ is the mass density of the medium, and Z and A are the atomic and mass numbers of the target gas. I is the mean excitation energy of the material.

For gaseous targets, the density varies with pressure under the ideal gas approximation, allowing definition of an effective thickness,

$$x_{\text{eff}} = x_{\text{ref}} \frac{P}{P_{\text{ref}}}, \quad (2)$$

where P is the measured gas pressure and x_{ref} , P_{ref} are reference values defined by the experimental geometry. Discrete mean excitation energies are obtained by applying numerical integration to the stopping power values extracted from the measured spectra at discrete effective thicknesses, using Simpson's method [3].

3. EXPERIMENTAL METHODOLOGY

The experiment was conducted using a gas chamber containing four alpha sources (^{244}Cm , ^{239}Pu , ^{241}Am , and ^{230}Th) and a silicon detector, as shown in Figure 1. The detector and sources are housed within the chamber, which is connected to a vacuum pump and gas supply system. Prior to measurements, the chamber is evacuated to ensure that only the selected gas is present. It is then filled with helium, nitrogen, or argon to controlled pressures.

For each gas, energy spectra are recorded at 18 different pressures. The alpha particles emitted from the four sources provide distinct energy peaks, allowing simultaneous measurements across multiple energies. The chamber pressure is monitored using a pressure gauge.

Signals from the detector are processed through a preamplifier and amplifier before being analysed by a multichannel analyser. The resulting spectra are recorded using Maestro software on a connected computer. This procedure is repeated for all pressures and gases to obtain the raw energy spectra.

Calibration data are acquired using a pulser with attenuation settings of 0.1, 0.2, 0.5, and 1. These data are used to determine the relationship between channel number and energy, as the recorded spectra are initially given in channel units [4].

4. DATA ANALYSIS

4.1. Peak Extraction Methodology

The extraction of peak channel values is performed using a multi-model fitting approach applied to both

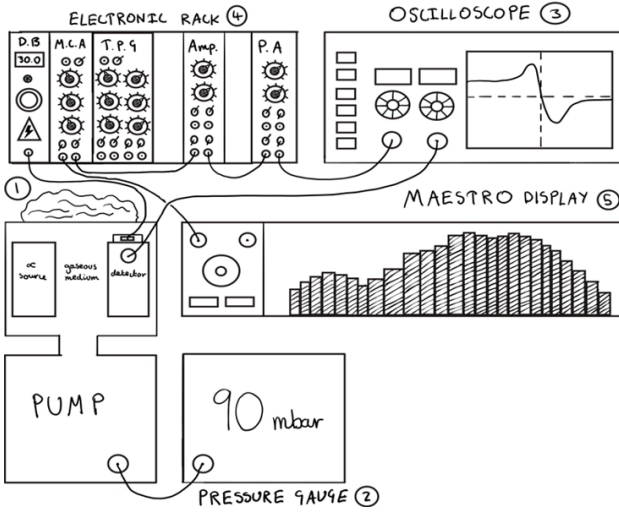


FIG. 1. Schematic of the experimental setup: (1) gas chamber containing the alpha source and silicon detector, connected to a vacuum pump; (2) pressure gauge; (3) oscilloscope; (4) electronics rack including bias supply, preamplifier, amplifier, multichannel analyser, and pulser; (5) computer running data acquisition software (Maestro).

calibration and experimental energy spectra. Each spectrum is fitted using six functional forms: Gaussian, generalised normal distribution, Crystal Ball function [5], Gaussian with a single-sided exponential tail, fourth-order polynomial, and a polynomial-exponential piecewise model spanning symmetric and asymmetric peak shapes to capture plausible detector responses and reduce model-dependent bias. For each model, an ensemble of fits is performed over a range of initial parameter values. Each fit is assessed using the reduced chi-squared statistic (χ^2_ν), and only fits satisfying a predefined acceptance window are re-

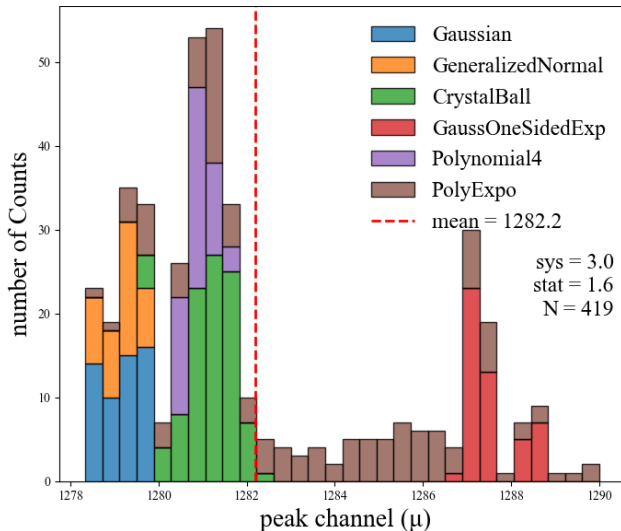


FIG. 2. Histogram of fitted peak channel positions (μ) showing counts for each peak position from the fitting procedure. Contributions from Gaussian (blue), generalized normal (orange), Crystal Ball (green), Gaussian with one-sided exponential (red), fourth-order polynomial (purple), and polynomial with exponential tail (brown) models are shown. A total of 419 fits are included, yielding $\mu = 1282.2 \pm 1.6$ (stat) ± 3.0 (sys).

tained. The accepted fits are used to construct a distribution of peak positions, as shown in Figure 2. This procedure ensures systematic sampling of statistical fluctuations and model dependence. The mean of this distribution is taken as the best estimate of the peak position, with the spread providing an estimate of model-dependent systematic uncertainty, while the statistical uncertainty is taken as the average uncertainty reported by the accepted fits.

This procedure is applied identically to all recorded spectra, including both calibration and experimental gas measurements, ensuring a consistent extraction of peak positions throughout the analysis.

4.2. Energy Calibration

The energy calibration establishes a relationship between detector channel number and energy using known alpha-particle energies [6]. Peak channel values are extracted from the calibration spectra using the procedure described in Section 4.1.

A linear fit is performed to determine the calibration parameters relating channel number to energy. The uncertainties on the peak positions are propagated through this fit to obtain uncertainties on the calibration parameters.

The resulting calibration is applied to all experimental spectra to convert channel values into physical energy units, enabling comparison across different pressures and gases.

4.3. Extraction of Mean Excitation Energy

For each gas and measured pressure, peak channel values are extracted for all sources using the methodology described in Section 4.1. The calibration parameters obtained in Section 4.2 are then applied to convert these channel values into energies.

The energy difference between successive measurements is used to determine the stopping power, corresponding to the left-hand side of Equation (1). The measured pressures are converted to effective thicknesses using Equation (2), where the reference pressure is taken as atmospheric pressure and the reference thickness corresponds to the length of the gas chamber.

The stopping power values are then used with Simpson's method to perform numerical integration, allowing the extraction of discrete mean excitation energies. This yields a set of I -values at different effective thicknesses for each source and gas.

For each gas, the I -values from all sources and thicknesses are combined, and an uncertainty-weighted average is used to determine the final mean excitation energy. The resulting distributions for all gases are shown in Figure 3.

5. RESULTS AND DISCUSSION

The extracted mean excitation energies for helium, nitrogen, and argon are summarised in Table I. The table presents the I -values obtained from each alpha source together with their corresponding uncertainties, as well as the final combined values for each gas. The quoted uncertainties for each data point include several contributions arising from the measurement and analysis procedure. These include uncertainties in the gas pressure, which propagate into the effective thickness, as well as uncertainties in the reference thickness. Additional contributions arise from density

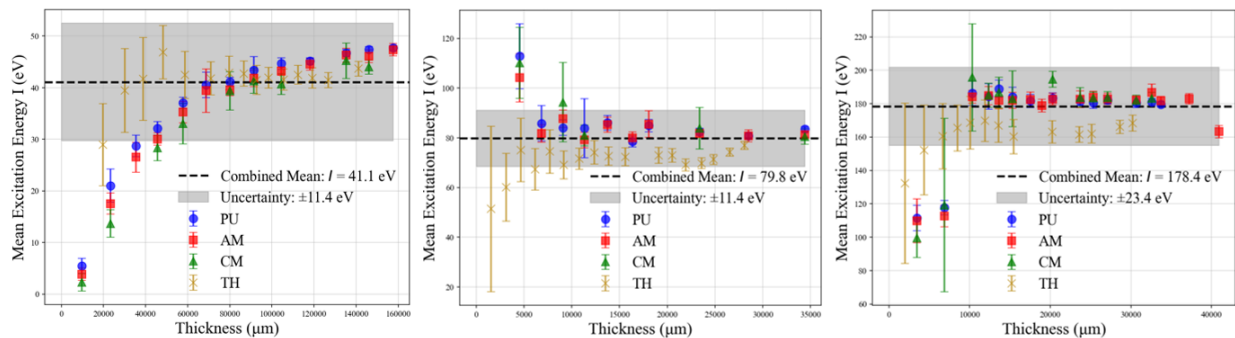


FIG. 3. Mean excitation energy as a function of effective gas thickness for helium (left), nitrogen (centre), and argon (right), where thickness is derived from measured gas pressure. Discrete I -values obtained from alpha sources ^{239}Pu (blue), ^{241}Am (red), ^{244}Cm (green), and ^{230}Th (yellow) are shown. The combined data yield mean excitation energies of 41.1 ± 11.4 eV for helium, 79.8 ± 11.4 eV for nitrogen, and 178.4 ± 23.4 eV for argon.

variations, the energy bounds used in the numerical integration, and the peak extraction and calibration procedures used to obtain these energies.

An additional uncertainty arises from the use of Simpson's rule in the numerical integration.

In addition, a systematic contribution is estimated from the spread of the I -value distributions obtained across repeated analyses. Uncertainty contributions are assumed independent and are combined in quadrature to obtain the total uncertainty for each data point.

A combined result for each gas is obtained by considering all data points from the four sources. The mean excitation energy is calculated using an uncertainty-weighted average over all values, while the systematic uncertainty is taken from the overall spread of the distribution.

For helium, a mean excitation energy of 41.1 ± 11.4 eV is obtained, in good agreement with the reference value of 41.8 ± 0.8 eV. For nitrogen, the measured value of 79.8 ± 11.4 eV is consistent with the reference value of 78 ± 7 eV. For argon, a value of 178.4 ± 23.4 eV is obtained, which is slightly lower than the reference value of 188 ± 10 eV, though still consistent within uncertainties [7].

6. CONCLUSION

The mean excitation energies of helium, nitrogen, and argon have been determined from alpha-particle energy loss measurements at varying gas pressures. The results are consistent with established reference values, indicating reliable determination of I -values within the quoted uncertainties.

The analysis extracts discrete I -values at different effective thicknesses rather than fitting a global stopping power curve. This preserves local structure in the data but leads to larger uncertainties compared

to a full-curve approach, which could reduce statistical uncertainty at the cost of increased model dependence.

The use of a uniform set of fitting models across all energy ranges also introduces limitations. A regime-dependent approach, applying different functional forms in low- and high-energy regions, could provide a more physically motivated description and improve systematic treatment.

Overall, the methodology yields consistent and physically reasonable mean excitation energies while highlighting clear routes for improving precision. The larger uncertainties, particularly for argon, arise from the spread in the extracted I -values and their propagation through numerical integration and experimental contributions.

TABLE I. Extracted mean excitation energies (I) for helium, nitrogen, and argon from individual alpha sources, together with the final combined values for each gas.

Gas	Source	I (eV)	Uncertainty (eV)	Final I (eV)
He	^{239}Pu	43.7	12.4	41.1 ± 11.4
	^{241}Am	37.4	13.0	
	^{244}Cm	31.1	14.9	
	^{230}Th	42.3	3.9	
	Combined			
N_2	^{239}Pu	82.4	9.6	79.8 ± 11.4
	^{241}Am	81.7	7.4	
	^{244}Cm	82.1	12.6	
	^{230}Th	72.9	6.2	
	Combined			
Ar	^{239}Pu	178.2	24.9	178.4 ± 23.4
	^{241}Am	179.8	23.9	
	^{244}Cm	182.9	30.5	
	^{230}Th	164.6	9.9	
	Combined			

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